Pseudo Ship-radiated Noise Generation Based on Adversarial Learning

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***Abstract*—The demand for acoustic positioning is increasing in various fields, meanwhile, it is very important to disturb acoustic positioning for underwater acoustics countermeasure and ship stealth technology. In this paper, a one-dimensional (1D) deep neural network based on adversarial learning to generate pseudo ship-radiated noises is presented, and a 1D convolutional network for classification is also given for evaluating the generated pseudo ship-radiated noises. The experimental results show that the proposed solution is effective to generate pseudo ship-radiated noises.**

Keywords—Ship-radiated Noise, Generative Adversarial Networks (GAN), Convolutional Neural Networks (CNN)

I. INTRODUCTION

The identification and classification of ship-radiated noises are militarily important to improve the effectiveness of underwater acoustics countermeasure. The ships could be identified and localized with the collected ship-radiated noises. Thus, the attack and defense strategies could be carried out [1] with the ship-radiated noises and underwater acoustic data generation technology. However, it is always difficult to obtain a large amount of the ship-radiated noises. Therefore, it is important to generate qualified ship-radiated noises in such scenario.

Some researchers have paid attention to the data generation by analyzing the true ship-radiated noises. The power spectra of ship-radiated noises are composed of continuous spectra and linear narrowband spectra at some discrete frequencies. Therefore, the traditional generation methods are to model the power spectra of ship-radiated noises, i.e, the continuous spectra [2-7], line spectra [2-6], and characteristics [4-5] of the ship-radiated noises are modeled separately. Most of these data generation techniques are not only costly and time-consuming, but also highly relied on *a priori* knowledge.

Even after the pseudo ship-radiated noises are generated, it is often a challenging task to evaluate them. The characteristics widely used for evaluation are onefold, consequently, the evaluation is not satisfactory. The nonlinear characteristics are also taken for identification [8-9]. However, professional knowledge is always required by these identification methods, and the computational complexity is also high. With the highly developing machine learning [10-14] and deep learning [15-17], some identification methods with higher computational efficiency and accuracy are presented. The spectral features of ship-radiated noise are extracted in [15-17], where the BP neural network [15-16] and deep neural network (DNN) [17] are taken to identify ship-radiated noises.

In this paper, the pseudo ship-radiated noise generation based on adversarial learning is addressed. A one-dimensional convolutional network for classification to evaluate the generated pseudo ship-radiated noises are also given, where the ship-radiated noises from the Shipsear dataset collected by Professor David in the University of Jago, Spain, are used for evaluation. Six kinds of ship radiated noises are included. The experimental results show the efficiency of the proposed one-dimensional ship-radiated noise generation network, and the generated pseudo ship-radiated noises could be better identified.

II. SHIP-RADIATED NOISE GENERATION AND CLASSIFICATION

A. The Framework

The framework of the proposed ship-radiated noise generation is illustrated in Fig. 1. The preprocessing module formats the collected ship-radiated noises for the deep neural network through normalization and sliding window. The classification module uses 1D convolutional neural networks to for ship-radiated noise identification. Based on the adversarial learning, the generation module is designed to generate pseudo noises radiated by different ships. The classification module is to evaluate the generated pseudo ship-radiated noises in seven aspects, i.e., waveform, spectra, and correlation, classification, etc.

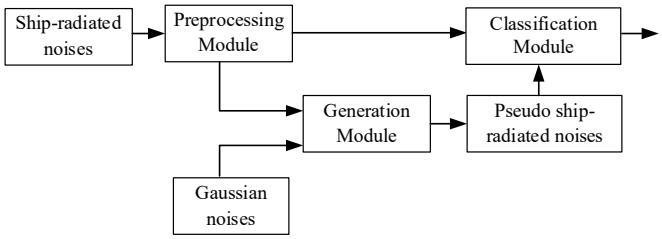


Fig. 1 The framework of the ship-radiated noise generation and verification.

B. Preprocessing Module

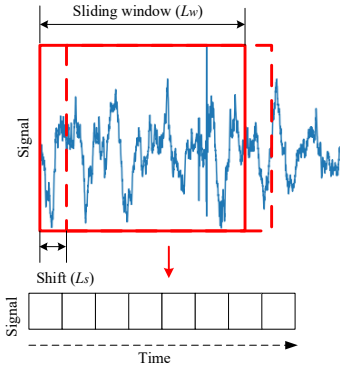


Fig. 2 The sliding windows and input segments.

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The collected ship-radiated noises are shown in Fig. 2. First, Min-Max standardization is used to normalize the data. Then the data are segmented by sliding windows [18]. The overlapped segments are obtained for the local temporal feature extraction. For the ship-radiated noise $x(t)$, the m-th input segment $\boldsymbol{x}^{(m)}$ is described as:

$$\boldsymbol{x}^{(m)} = \left[x\big((m-1)\times L_s+1\big)\cdots x\big((m-1)\times L_s+L_w\big)\right] \quad (1)$$

where L_w and L_s are the sliding window length and step size, respectively.

C. Classification Module

Classification module is used to evaluate the quality of the generated pseudo ship-radiated noises. The input to the module is ship-radiated noise and outputs are the probabilities corresponding to different kinds of noises. The 1D convolutional network for 1D ship-radiated noise classification is composed of four 1D convolutional layers, two 1D max-pooling layers, one flatten layer, one fully connected hidden layer and one output layer. The 1D convolutional layer extracts relevant features from ship-radiated noises, the 1D pooling layer reduces model parameters by down-sampling and prevents overfitting; the fully connected layer and the output layer are used for classification.

D. Generation Module

The generation module for generating 1D pseudo ship-radiated noises is shown in Fig. 3, which consists of two 'adversarial' models, i.e, a generative model/generator (G) and a discriminative model/discriminator (D).

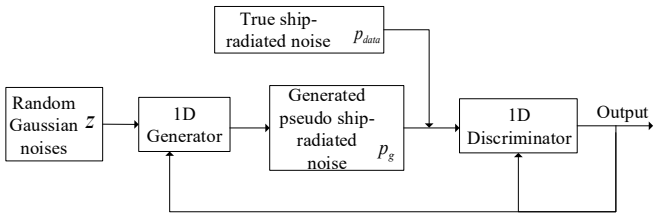


Fig. 3 Generation module

(1) Generator

The input to the generator is random Gaussian noise z, and output the generated pseudo ship-radiated noise. The generator G is shown in Fig. 4, which consists of [Fully connected layer] →[Reshape layer] →[1D upsampling layer 1] → [1D convolutional layer 1] → [1D upsampling layer 2] →[1D convolutional layer 2] →[1D convolutional layer 3].

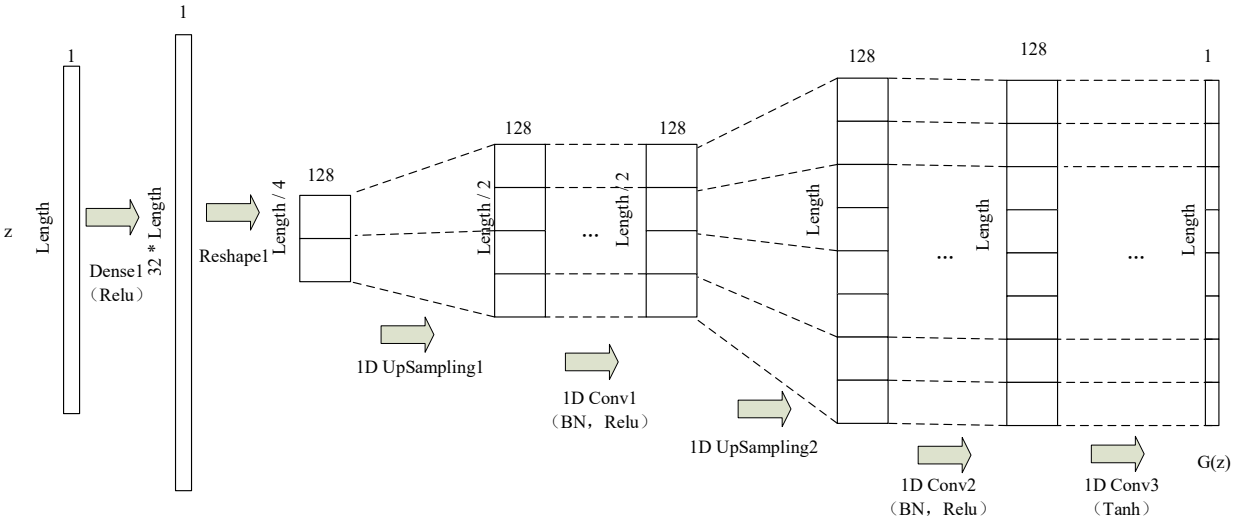


Fig. 4 The generator (G)

(2) Discriminator

The input to the discriminator is the generated pseudo ship-radiated noise or true ship-radiated noise, and output the probability that the ship-radiated noise is true. The discriminator is shown in Fig. 5, which uses [1D convolutional layer 1] to extract relevant features. Then the extracted relevant features are input into two branches of deep neural networks, respectively.

The first branch of deep neural networks consists of [1D convolutional layer 2] → [1D ZeroPadding layer] → [1D convolutional layer 3] →[1D convolutional layer 4] →[Flatten layer] →[Output Layer 1]. The second branch of deep neural networks consists of [long short-term memory (LSTM) layer 1] →[LSTM layer 2] →[LSTM layer 3] →[Output layer 2].

These two branches of deep neural networks are trained separately. Then we combine these two branches of deep neural networks according to the multiplication rule and obtain the final discriminate results, which means the discriminator would make the correct classification only when both two branches of deep neural networks well trained. Thus, the accuracy of the discriminator would be effectively improved.

(3) Loss function

The generator G and discriminator D are trained at the same time, and both follow the Minimax game. Therefore, the loss function of the 1D deep neural network to generate ship-radiated noise is given by:

$$\min_G \max_D V(D,G) = E_{x\sim P_{data(x)}}[log\, D(\boldsymbol{x})] + E_{\boldsymbol{z}\sim p_z(\boldsymbol{z})}\left[log\left(1-D\big(G(\boldsymbol{z})\big)\right)\right] \quad (2)$$

where $\boldsymbol{z}$ is the Gaussian noise, P_z is the distribution function of the Gaussian noise, and P_{data} is the distribution function of true ship-radiated noises.

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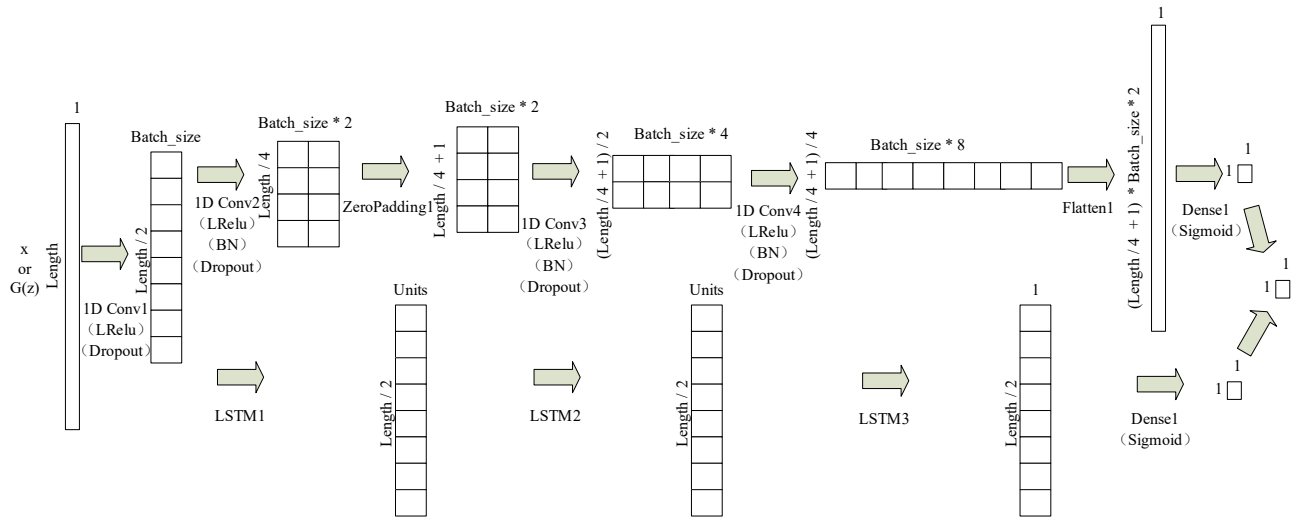


Fig. 5 The discriminator (D)

III. Experimental Results

A. Experiment Description

The ship-radiated noises are taken to verify the effectiveness of the proposed solution for the ship-radiated noise generation. The ship-radiated noises are from the Shipsear dataset, which are collected by self-contained hydrophones. The sampling frequency is 52734Hz, every segment has 128 samples, i.e, $L_w = 128$. The noises are radiated by six kinds of ships, i.e., Arroios, Bateero, Costa Voyager, Mar de Mouro, Pirata de Cies, and Pirata de Salvora. The abbreviations and data lengths of six kinds of ship-radiated noises are described in Table 1.

Table 1 Abbreviation and data lengths of six kinds of ship radiated noises

Abbreviation	Size	Description
Data_1	(11411526,1)	Noises collected from Arroios
Data_2	(16562653,1)	Noises collected from Bateero
Data_3	(21129151,1)	Noises collected from Costa Voyager
Data_4	(13183500,1)	Noises collected from Mar de Mouro
Data_5	(10940823,1)	Noises collected from Pirata de Cies
Data_6	(8490174,1)	Noises collected from Pirata de Salvora

The waveforms of six kinds of ship-radiated noises are illustrated in Fig. 6.

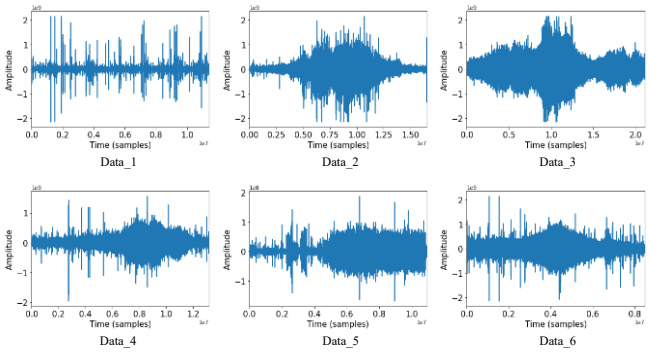


Fig. 6 The waveforms of six kinds of ship-radiated noises

For generation module, in the process of training, the batch size is 32 and the optimizer for both the generative model and the discriminative model is the Adam optimizer, with the learning rate set to 0.0002, and the other parameters as the default. For classification module, the batch size is 128, and Adam with default parameters is employed.

B. Analysis and Comparison

The comparison between the true and generated pseudo ship-radiated noises in seven aspects, e.g., waveform, power spectra, cepstra, principal component analysis (PCA), autocorrelation (ACF), partial autocorrelation (PACF) and classification accuracy, which are generally used to identify the ship-radiated noises by the traditional methods. Due to space limitations, only the comparison on Data_1 to Data_3 are illustrated in this paper.

The waveforms, power spectral density (PSD) are shown in Fig. 7 and Fig. 8, respectively. It is shown the waveforms and PSD of generated pseudo ship-radiated noises are highly similar to those of the true ship-radiated noises.

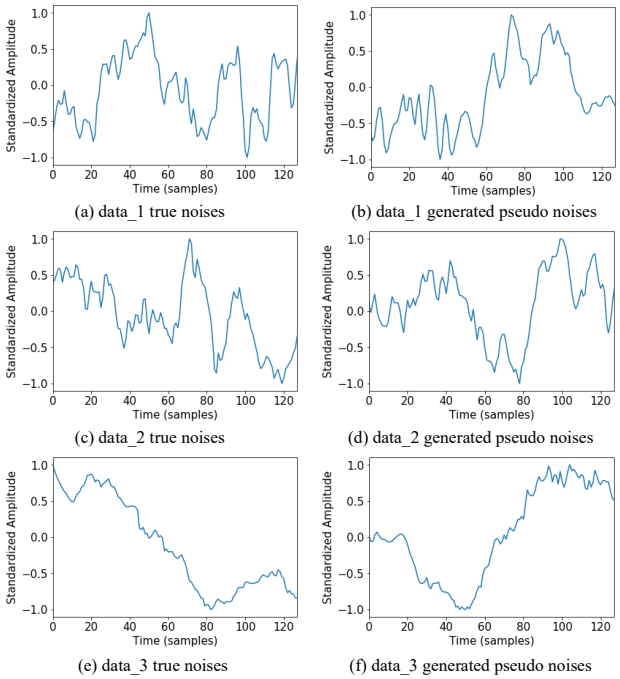
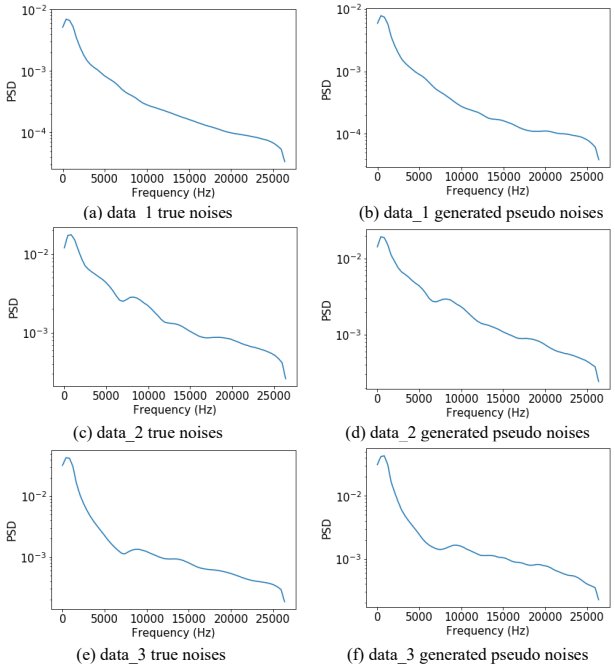


Fig. 7 The waveforms of (left) true and (right) generated pseudo ship-radiated noises

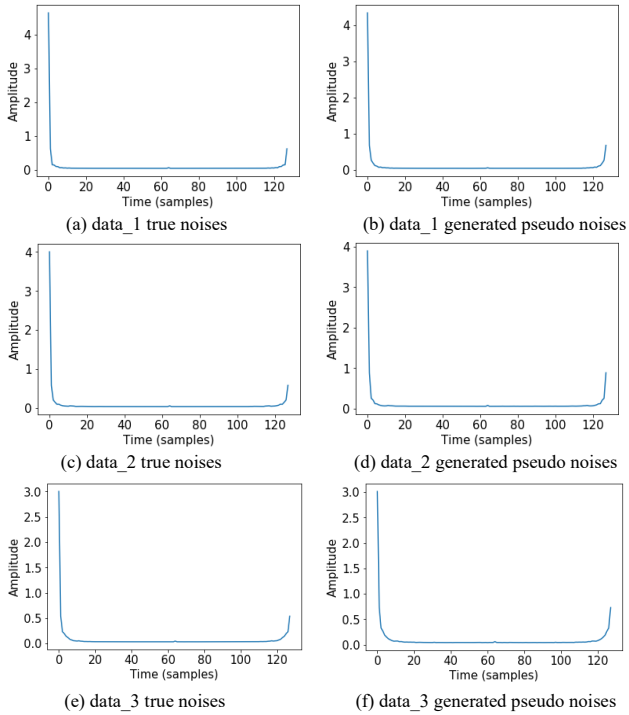
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(a) data_1 true noises (b) data_1 generated pseudo noises
(c) data_2 true noises (d) data_2 generated pseudo noises
(e) data_3 true noises (f) data_3 generated pseudo noises

Fig. 8 The PSD of (left) true and (right) generated pseudo ship-radiated noises

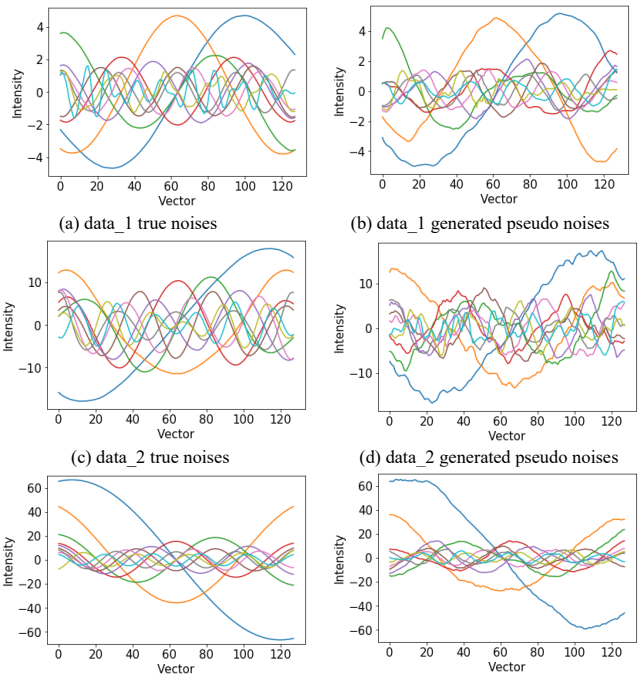
The cepstra could show the periodic structure of complex spectra, and thus extract the periodic components. Fig.9 illustrates the cepstra of the true noises and generated pseudo ship-radiated noises in a consistent manner.



(a) data_1 true noises (b) data_1 generated pseudo noises
(c) data_2 true noises (d) data_2 generated pseudo noises
(e) data_3 true noises (f) data_3 generated pseudo noises

Fig. 9 The cepstra of (left) true and (right) the generated pseudo ship-radiated noises

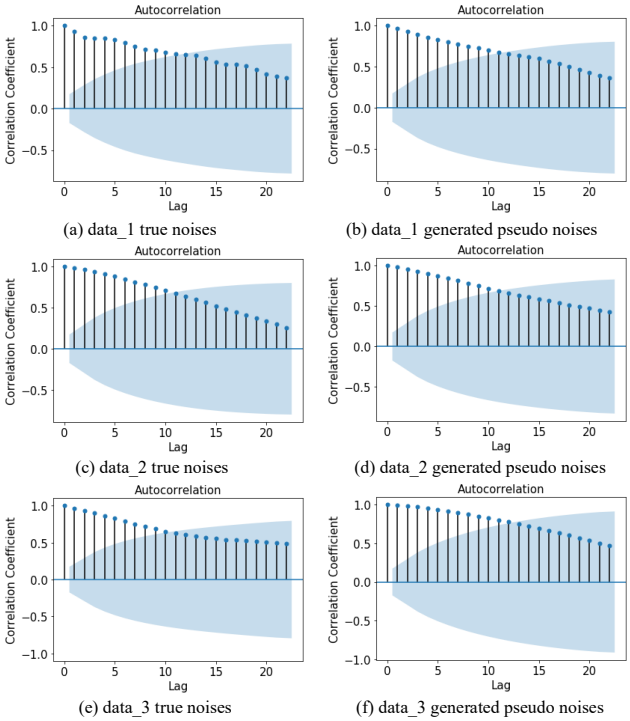
First 10 principal components of true and generated pseudo ship-radiated noises are shown in Fig. 10, which means the low-rank components of generated pseudo ship-radiated noises are also similar to those of true ship-radiated noises.



(a) data_1 true noises (b) data_1 generated pseudo noises
(c) data_2 true noises (d) data_2 generated pseudo noises
(e) data_3 true noises (f) data_3 generated pseudo noises

Fig. 10 The PCA of (left) true and (right) generated pseudo ship-radiated noises.

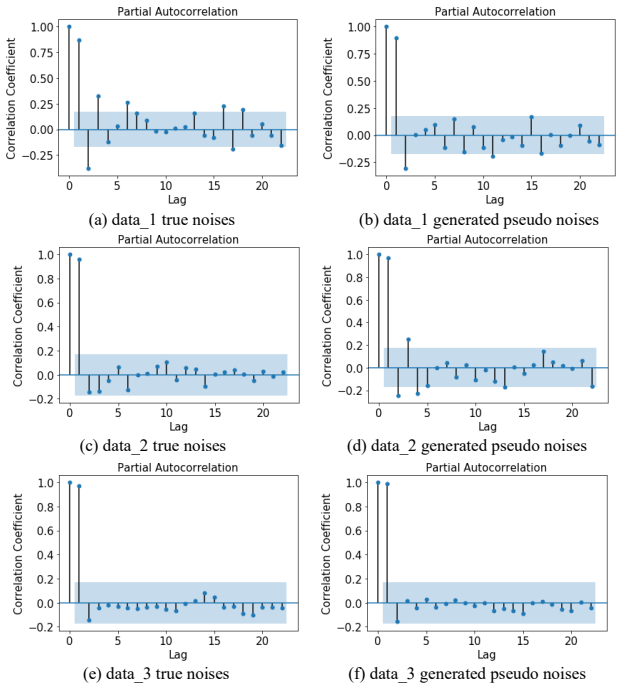
Autocorrelation and partial autocorrelation are shown in Fig. 11 and 12, which indicate the true and generated pseudo ship-radiated noises have a similar relationship in the time domains.



(a) data_1 true noises (b) data_1 generated pseudo noises
(c) data_2 true noises (d) data_2 generated pseudo noises
(e) data_3 true noises (f) data_3 generated pseudo noises

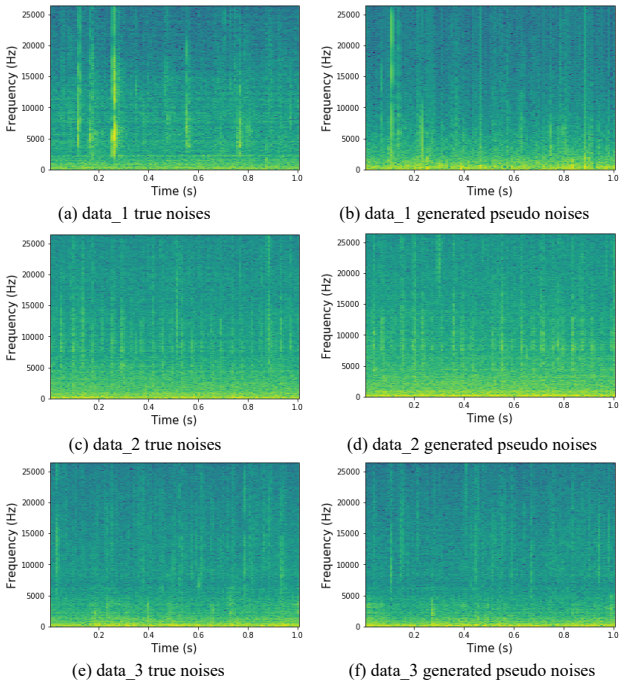
Fig. 11 The autocorrelation of (left) true and (right) generated pseudo ship-radiated noises

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(a) data_1 true noises (b) data_1 generated pseudo noises
(c) data_2 true noises (d) data_2 generated pseudo noises
(e) data_3 true noises (f) data_3 generated pseudo noises

Fig. 12 The partial autocorrelation of (left) true and (right) generated pseudo ship-radiated noises



(a) data_1 true noises (b) data_1 generated pseudo noises
(c) data_2 true noises (d) data_2 generated pseudo noises
(e) data_3 true noises (f) data_3 generated pseudo noises

Fig. 13 The time-frequency representations of (left) true and (right) the infinite length generated pseudo ship-radiated noises.

A 1D convolutional neural network is trained by the true ship-radiated noises for classification and the classification accuracy is shown in Table 2. It is shown the generated ship-radiated noises are effectively classified, which means the classification model trained by the true ship-radiated noises could not discriminate between the true and pseudo ship-radiated noises.

Table 2 The classification accuracy of (top) true noises and (bottom) the generated pseudo ship-radiated noises

	Data_1	Data_2	Data_3
True noises	0.929	0.761	0.906
Generated pseudo noises	0.972	0.717	0.988

C. *Infinite Length Pseudo Ship-radiated Noise Generation*

The classification accuracies of true and infinite length pseudo ship-radiated noises are shown in Table 3. The experimental results show that the pseudo noise fragments of each kind of ship are spliced smoothly, and there is no obvious breakpoint at the splicing place. We realize the generation of infinite length ship pseudo noise data.

Table 3 The classification accuracies of (top) true and (bottom) the infinite length pseudo ship-radiated noises

	Data_1	Data_2	Data_3
	True ship-radiated noises		
Training	0.928	0.765	0.910
Testing	0.929	0.761	0.906
	Generated pseudo ship-radiated noises		
Segments	0.972	0.717	0.988
Infinite length	0.974	0.689	0.979

Table 4 The classification accuracies in the time-frequency domain of (top) true and (bottom) the infinite length generated pseudo ship-radiated noises

	Data_1	Data_2	Data_3
	True ship-radiated noises		
Training	0.914	0.776	0.967
Testing	0.916	0.773	0.966
	Generated pseudo ship-radiated noises		
Segments	0.939	0.787	0.980
Infinite length	0.939	0.717	0.981

The time-frequency representations of true and infinite length generated pseudo ship-radiated noises are shown in Fig.13, and the classification accuracies in the time-frequency domain are shown in Table 4. The experimental results show the infinite length generated pseudo ship-radiated noises are smoothly spliced.

IV. Conclusions

In this paper, a 1D data generation network based on adversarial learning has been presented for generating pseudo ship-radiated noises, and a 1D convolutional network for classification has been also given to evaluate the generated pseudo ship-radiated noises. The experimental results show the pseudo ship-radiated noises could be successfully generated by the proposed 1D data generation network.

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